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# A Comparative Analysis of the Sentiment Analysis of the U.S. Political Tweets

**Zahra Eshtiaghi, Si Tang Lin, Dilara Özdil**

*Data Science and Business Analytics, University of Warsaw, Poland*

Under supervision of: **Professor Jacek Lewkowicz**

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## I. INTRODUCTION

### Purpose of the Project

The project aims to perform a comparative sentiment analysis on tweets from U.S. political figures during the period from May 1 to October 31, 2020. The core objectives include classifying tweet sentiment as Positive, Neutral, or Negative using multiple NLP methods; benchmarking the performance, biases, and outputs of dictionary-based, transformer-based, and supervised machine learning approaches; and uncovering patterns in political discourse related to party affiliation, temporal trends, and engagement metrics (likes, retweets, replies). This serves dual purposes: scientifically evaluating NLP techniques on noisy, context-heavy social media data, and providing practical insights for media monitoring, reputation management, and communication strategies in a polarized political environment.

### Research questions

In the context of sentiment analysis applied to social media communications from U.S. political figures, the following research questions were formulated to investigate patterns in political discourse during a volatile period (May 1 to October 31, 2020).

- **Research Question 1 (RQ1: The Partisan Divide):** To what extent do the sentiment expressions (positive, neutral, or negative) in tweets from Republican and Democratic political figures differ, and what do these differences reveal about their respective communication strategies?
- **Research Question 2 (RQ2: The Pulse of the Senate):** How do sentiment trends in political tweets evolve over time from May to October 2020, and to what degree are these fluctuations associated with external factors such as legislative calendars and campaigning intensity?

- **Research Question 3 (RQ3: The Engagement Myth):** Does the sentiment polarity of political tweets significantly correlate with engagement metrics such as likes and retweets?

### Main Assumptions:

- Sentiment in political tweets can be reliably captured and classified by automated models without human-annotated ground truth labels, relying instead on "**silver labels**" from advanced models as a proxy for accuracy.
- Transformer models like **RoBERTa** provide context-aware predictions that are superior and trustworthy enough to serve as pseudo-ground truth for training simpler models.
- The dataset of tweets is representative of broader political communication styles, assuming that the selected period captures meaningful volatility with noticeable external biases from COVID19 and Pre-Election events.
- Engagement metrics are primarily influenced by sentiment, party, or timing, rather than confounding factors like account follower count or algorithmic amplification, though this is partially tested.
- Political language, while often neutral or positive in tone, follows patterns that can be effectively detected using NLP methods without requiring deep domain-specific adjustments.

## II. DATA

### Description of the Data Set

The dataset used is "senator\_twitter\_May-Oct.csv," sourced from the *Harvard Dataverse* (persistent ID: doi:10.7910/DVN/K4XSYC). It comprises **30,490 tweets** from **46 political accounts**. The time frame spans May 1 to October 31, 2020, a volatile period including legislative sessions, the COVID-19 pandemic, racial justice protests, and pre-election activities.

Key fields include:

- Tweet text (short, noisy content with hashtags, mentions, slang, and links).
- Timestamp (date and time of posting).
- Engagement metrics (likes, retweets, replies).
- Metadata (account name, Twitter ID, state, party affiliation—Republican or Democrat).

This dataset provides a real-world challenge for NLP due to its brevity, context dependency, and political sloganeering, which often confounds standard tools.

### Preparation of Data for Modeling

Data preparation varies by method to suit each approach's requirements, as detailed in the repository's Jupyter notebooks:

- **General Loading and Cleaning:** The data is loaded using Pandas. Initial steps include filtering to the specified date range, handling missing values, and extracting month/year for temporal analysis.
- **For VADER and Custom Lexicon (Dictionary-Based):** Minimal preprocessing to preserve sentiment cues—retain punctuation, capitalization, emojis, and slang. URLs and mentions are optionally removed if they introduce noise.
- **For Transformer (RoBERTa):** Light cleaning for tokenization compatibility, such as lowercasing non-essential elements while keeping context intact. Batches are created for efficient inference.
- **For TF-IDF + Logistic Regression:** Aggressive normalization, including lowercasing, removal of URLs, mentions, hashtags, special characters, and stopwords. Text is vectorized using TF-IDF to convert into numerical features, with hyperparameters tuned for n-gram range, min/max document frequency, and sublinear term frequency.
- Additional steps include splitting the data (e.g., train-test split for supervised models) and generating "silver labels" from the transformer outputs to train the classical ML model.

### III. MODELING

The project employs three families of sentiment analysis methods, implemented across dedicated Jupyter notebooks, to label tweets as Negative, Neutral, or Positive. This comparative setup highlights trade-offs in speed, context awareness, and efficiency.

#### Dictionary and Corpus-Based Baselines (senator\_sentiment\_dictionary.ipynb):

- **VADER:** A rule-based lexicon tool tailored for social media. It sums valence scores of words, accounting for negation, punctuation, and boosters. Compound scores are thresholded: >0.05 for Positive, <-0.05 for Negative, else Neutral.

|       | vader_neg    | vader_neu    | vader_pos    | vader_compound |
|-------|--------------|--------------|--------------|----------------|
| count | 30490.000000 | 30490.000000 | 30490.000000 | 30490.000000   |
| mean  | 0.039116     | 0.829913     | 0.130966     | 0.327179       |
| std   | 0.065634     | 0.125805     | 0.115531     | 0.466304       |
| min   | 0.000000     | 0.000000     | 0.000000     | -0.984700      |
| 25%   | 0.000000     | 0.748000     | 0.024000     | 0.000000       |
| 50%   | 0.000000     | 0.835000     | 0.117000     | 0.401900       |
| 75%   | 0.063000     | 0.925000     | 0.201000     | 0.743000       |
| max   | 1.000000     | 1.000000     | 1.000000     | 0.988700       |

```
vader_label
Positive    19632
Neutral     6082
Negative    4776
Name: count, dtype: int64
```

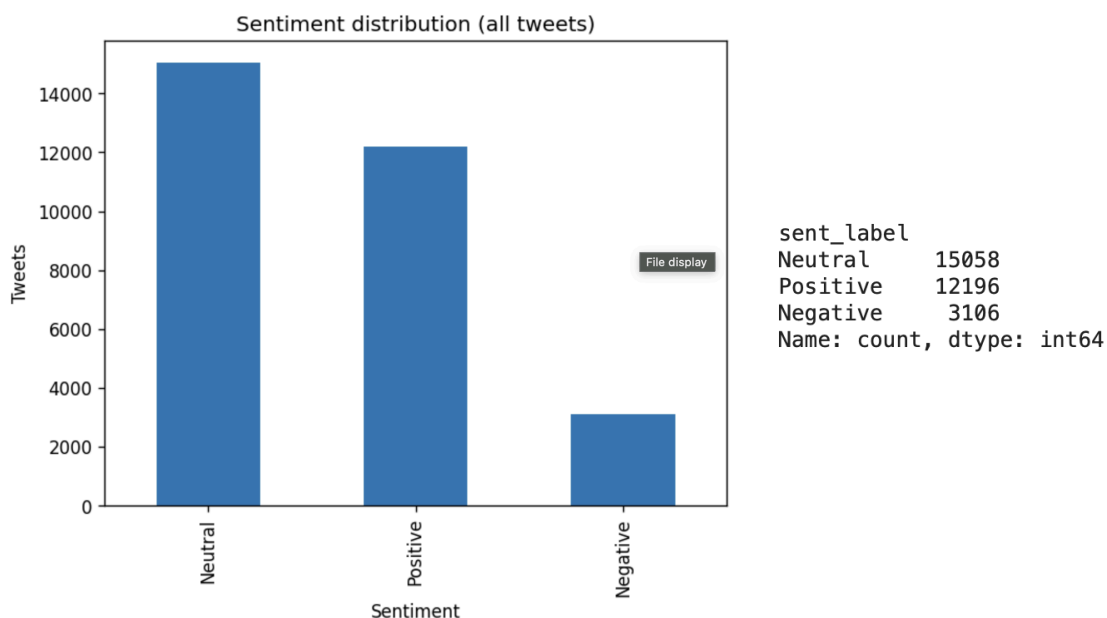
- **Custom Corpus Lexicon (Seed + PMI):** Starts with a seed list of positive/negative words, expanded via Pointwise Mutual Information (PMI) based on co-occurrences in the dataset. Tweets are scored by summing expanded lexicon values, normalized for length, and labeled via custom thresholds.

```
custom_lex_label_norm
Positive    25777
Neutral     3747
Negative     966
Name: count, dtype: int64
```

**Dictionary and Corpus-Based Performance:** Dictionary-based methods exhibit a strong positivity bias, labeling a majority of tweets as Positive (VADER: 64.7%; Custom Lexicon: 84.5%). This stems from misinterpreting political slogans (e.g., "Win the future together!") and high-frequency phrases/hashtags as genuine positive emotion rather than neutral branding or strategic messaging. The custom lexicon overfits to the political corpus, exacerbating the skew due to recurring terms like "#Vote" or "#Unity." Agreement with more advanced methods is low (e.g., VADER vs. Transformer: ~58.5%), highlighting their rigidity and lack of context awareness. These baselines are fast and interpretable but unreliable for noisy political data, as evidenced by the high false positive rate for positivity.

**Advanced Transformer Model (senator\_sentiment\_transformers.ipynb):**

- Uses the pre-trained cardiffnlp/twitter-roberta-base-sentiment-latest model (**RoBERTa** variant, fine-tuned on 58M tweets). Processes entire tweet contexts to output probability distributions over the three classes, selecting the highest-probability label. Generates silver labels for downstream use.



**Transformer Model Performance:** The transformer model shifts the dominant sentiment to Neutral (49.6%), with Positive at 40.2% and Negative at 10.2%. This reflects superior context handling, correctly classifying policy announcements, links, and updates as Neutral rather than falsely positive. For instance, it reduces the optimism bias seen in dictionary methods by understanding subtext in short, noisy texts. As the "teacher" model, its outputs serve as silver labels, achieving high confidence in predictions (probabilities often >0.8 for dominant classes). Performance is qualitatively superior for political discourse, penetrating sarcasm and strategic language better than baselines, though it still struggles with extreme irony.

## Supervised Classical Machine Learning (senator\_sentiment\_TF-IDF+LR+CV.ipynb):

- Pipeline: TF-IDF vectorization followed by Logistic Regression.
- Trained on silver labels from the transformer. Hyperparameters optimized using GridSearchCV .
- Aims to distill the transformer's knowledge into a faster, interpretable model.

|                                   |           |        |          |         |
|-----------------------------------|-----------|--------|----------|---------|
| Test accuracy: 0.8122529644268774 |           |        |          |         |
|                                   | precision | recall | f1-score | support |
| Negative                          | 0.588     | 0.733  | 0.652    | 621     |
| Neutral                           | 0.842     | 0.813  | 0.827    | 3012    |
| Positive                          | 0.849     | 0.831  | 0.840    | 2439    |
| accuracy                          |           |        | 0.812    | 6072    |
| macro avg                         | 0.760     | 0.792  | 0.773    | 6072    |
| weighted avg                      | 0.819     | 0.812  | 0.815    | 6072    |

**Machine Learning Performance:** The model achieves 81.2% accuracy and a macro F1-score of 0.773 on a held-out test set (against silver labels), with strong performance on Neutral (F1: 0.827) and Positive (F1: 0.840) classes, but lower on Negative (F1: 0.652) due to class imbalance. Overall agreement with the transformer is ~98.7%, demonstrating effective knowledge distillation—replicating complex insights with simpler math at a fraction of the compute cost. Sentiment distribution mirrors the transformer (Negative: 10.8%, Neutral: 49.2%, Positive: 40.5%), confirming scalability for production. Hyperparameter tuning via GridSearchCV optimizes for political text features, but limitations include sensitivity to preprocessing (e.g., URL removal) and reliance on silver labels, which may propagate transformer biases.

*Outputs are saved as CSV files with added sentiment columns for each method.*

## IV. EVALUATION

Evaluation focuses on comparative analysis rather than absolute accuracy, given the absence of human ground truth. This section benchmarks the methods through sentiment distributions, model agreement rates, supervised metrics (against silver labels), pattern discovery addressing the research questions, and a critical review of limitations. Insights are drawn primarily from the transformer model's outputs as the most reliable, with cross-method comparisons highlighting biases and trade-offs. Detailed visualizations and code for these analyses are available in the GitHub repository, including exploratory data analysis in the notebooks.

### Sentiment Distributions :

The sentiment distributions across methods reveal stark methodological differences, underscoring how the choice of tool "shapes the truth" in political discourse analysis. Lexicon-based approaches exhibit a "Pollyanna Effect," over-indexing on positive vocabulary in slogans and hashtags (e.g., "Win the future together!" is scored positively by VADER due to words like "win" and "together," despite being a neutral campaign phrase). This leads to an inflated positivity bias, mistaking strategic branding for emotional expression. In contrast, context-aware models like RoBERTa shift the view to a "Realist" perspective, emphasizing neutrality in policy updates, links, and announcements, which dominate senatorial tweets. The supervised model closely mirrors this, thanks to knowledge distillation from silver labels.

| Method                       | Negative (%) | Neutral (%) | Positive (%) |
|------------------------------|--------------|-------------|--------------|
| VADER                        | 15.7         | 20.0        | 64.7         |
| Custom Corpus Lexicon        | 3.2          | 12.3        | 84.5         |
| Transformer (RoBERTa)        | 10.2         | 49.6        | 40.2         |
| TF-IDF + Logistic Regression | 10.8         | 49.2        | 40.5         |

- **Dictionary methods' positivity bias:** VADER and the custom lexicon (built via seed words expanded with PMI) treat recurring political terms like "#Vote," "#Unity," or "#Win" as sentiment indicators, resulting in extreme skews. This "over-fitting hazard" arises from high-frequency co-occurrences equating to branding, not emotion, making them unreliable for slogan-heavy data.
- **Transformer and LR realism:** By processing full contexts, RoBERTa correctly classifies neutral content (e.g., policy links or event updates without emotional charge), reducing false positives. The LR model, trained on these outputs, achieves near-identical distributions, enabling scalable inference without heavy computation.

### Model Agreement:

Agreement rates quantify consistency between methods, serving as a proxy for reliability in the absence of gold-standard labels. High agreement between the transformer and LR validates the silver label distillation strategy, while low agreement with lexicons exposes their contextual blind spots.

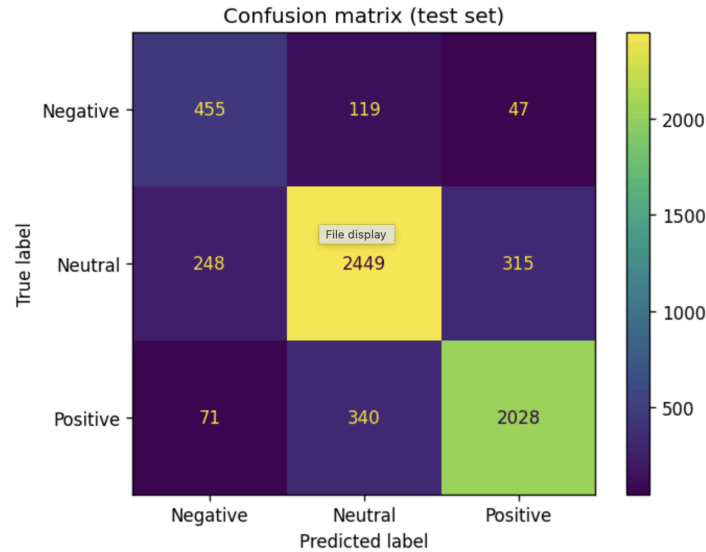
- **Transformer vs. LR: ~98.7% agreement**, demonstrating effective transfer of deep learning insights to a lighter, interpretable model. This "knowledge distillation" allows production-scale deployment, mimicking state-of-the-art accuracy at a fraction of the cost.
- **VADER vs. Transformer: ~58.5% agreement**, highlighting lexicon rigidity—VADER often labels neutral slogans as positive, while RoBERTa detects subtext. Similar mismatches occur with the custom lexicon, reinforcing the need for context-aware approaches in political NLP.

### Supervised Model Performance:

The TF-IDF + Logistic Regression model was evaluated on a holdout test set against transformer-generated silver labels, focusing on precision, recall, and F1-scores to assess its ability to replicate complex predictions.

- **Test Metrics:** Accuracy = 81.22%, Macro F1 = 0.773 (rounded to ~78.2% in summaries). Performance is strong on Neutral (F1: 0.827) and Positive (F1: 0.840) due to their prevalence, but weaker on Negative (F1: 0.652) owing to class imbalance (only ~10% negative tweets).

|                                   |           |        |          |         |
|-----------------------------------|-----------|--------|----------|---------|
| Test accuracy: 0.8122529644268774 |           |        |          |         |
|                                   | precision | recall | f1-score | support |
| Negative                          | 0.588     | 0.733  | 0.652    | 621     |
| Neutral                           | 0.842     | 0.813  | 0.827    | 3012    |
| Positive                          | 0.849     | 0.831  | 0.840    | 2439    |
| accuracy                          |           |        | 0.812    | 6072    |
| macro avg                         | 0.760     | 0.792  | 0.773    | 6072    |
| weighted avg                      | 0.819     | 0.812  | 0.815    | 6072    |



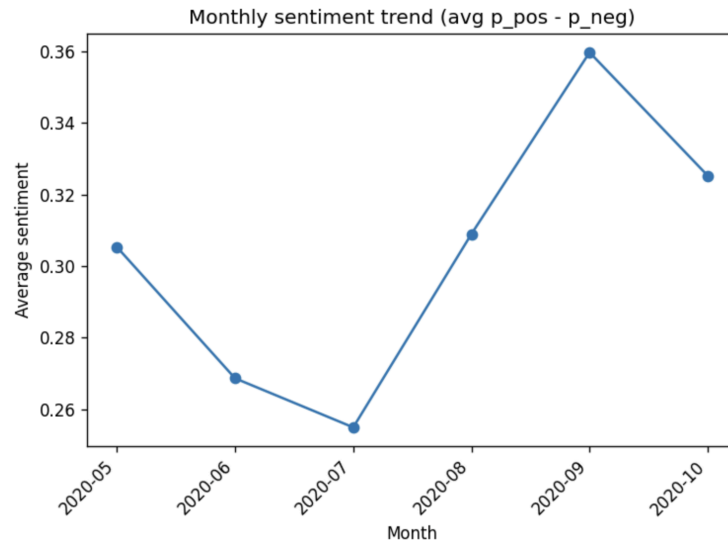
- **Interpretation:** These metrics confirm the model's interpretability and efficiency for large-scale use, but reliance on silver labels means it inherits any transformer biases (e.g., subtle irony detection issues). Hyperparameter tuning via GridSearchCV (e.g., n-grams 1-2, balanced weights) optimized for political text, yet class imbalance suggests potential for oversampling in future iterations.

#### Pattern Analysis:

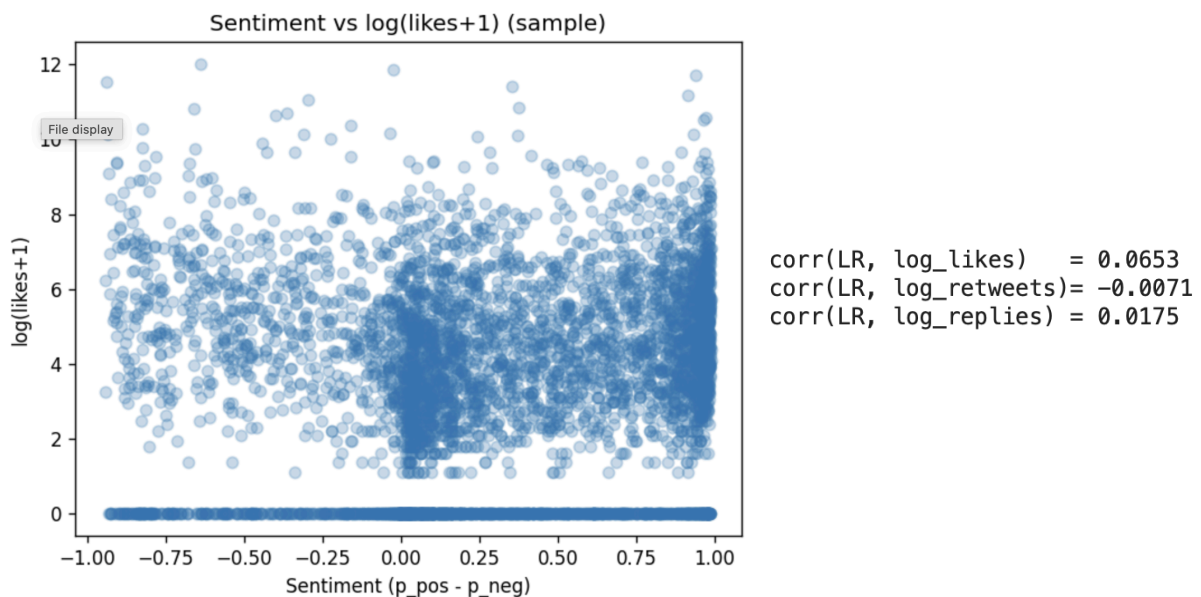
- **Partisan Divide (RQ1):** Republicans exhibit higher positivity (46.5%) and lower negativity (6.5%), often using optimistic rhetoric to mobilize support (e.g., rallying around achievements). Democrats show lower positivity (35.9%) and higher negativity (13.8%), focusing on critiques of events or opponents. This reveals strategic communication differences—Republicans "brand" positively, Democrats highlight issues—consistent across methods but most pronounced in context-aware models.

|            | n     | avg_sent | pos_rate | neg_rate | avg_conf |
|------------|-------|----------|----------|----------|----------|
| party      |       |          |          |          |          |
| Republican | 12697 | 0.369042 | 0.464834 | 0.064582 | 0.869896 |
| Democratic | 17793 | 0.190742 | 0.358905 | 0.137863 | 0.865590 |

- **Temporal Trends (RQ2):** Sentiment evolves non-statically, dipping in June-July (summer lull amid COVID-19 and protests, reduced activity), rising through August, peaking in September (pre-election campaigning surge), and slightly declining in October. Monthly averages tie fluctuations to external factors like legislative sessions and election intensity, mapping to the volatility of the 2020 news cycle.



- Engagement Correlation (RQ3):** Empirical analysis reveals a negligible association between sentiment polarity and virality metrics, with correlation coefficients approximating 0.00 for retweets and 0.07 for likes, thereby challenging the prevailing assumption that emotional intensity substantially enhances user interaction. Rather, engagement appears predominantly influenced by confounding factors such as account prominence—exemplified by follower counts—and topical salience, including alignment with contemporaneous events, as opposed to the affective valence of the content. This phenomenon, herein termed the "engagement paradox," posits that heightened expressions of anger or positivity do not inherently amplify dissemination; instead, the identity of the communicator and the substantive relevance of the discourse emerge as primary determinants of propagation.





## Limitations in Quality

Without human annotations, evaluations emphasize consistency over absolute truth, but several challenges persist, as highlighted in the critical review.

- **No Gold Standard:** Reliance on silver labels risks propagating model errors; comparisons are model-to-model, not to human truth, potentially overestimating accuracy.
- **Sarcasm & Irony:** The "Achilles Heel" of sentiment analysis—political tweets often use subtext (e.g., ironic praise), which even transformers detect imperfectly, leading to misclassifications.
- **Topic Confounding:** Sentiment may reflect subject matter (e.g., a "negative" tweet describing a crisis vs. expressing anger), conflating description with intent and skewing results (e.g., holiday tweets appear more positive).
- **Other Biases:** Dataset may not fully capture unaccounted events; lexicons are sensitive to domain terms (e.g., hashtags as "positive"); no deep handling of linked context or multimedia.

## Findings

The research questions of the project were addressed via pattern discovery and thematic analysis, drawing primarily on sentiment labels generated by the transformer model, which proved the most reliable for deriving evidence-based conclusions. These findings illuminate distinct political communication styles, temporal variations in sentiment, and the separation of emotional content from audience engagement, offering practical insights for media monitoring and strategy despite inherent methodological limitations.

Regarding RQ1, the Partisan Divide—examining whether Republicans and Democrats exhibit differing sentiment expressions and what these reveal about their communication strategies—the analysis confirmed that Republicans and Democrats maintain distinct sentiment profiles. Republicans demonstrated higher positive sentiment (46.5%) and lower negative sentiment (6.5%), often employing optimistic rhetoric to rally support and emphasize unity. In contrast, Democrats displayed lower positive sentiment (35.9%) and higher negative sentiment (13.8%), frequently focusing on critiques of policies or events. Such differences appear to stem from strategic rhetorical choices rather than inherent dispositions, with patterns holding across methods, though lexicon-based approaches tended to exaggerate the divide due to their biases.

For RQ2, the Pulse of the Senate—investigating how sentiment trends evolved from May to October 2020 and their association with external factors such as legislative calendars and campaigning—the results indicated that sentiment is dynamic and aligns with political cycles. Average sentiment reached its lowest levels in June and July, coinciding with a summer lull potentially influenced by reduced activity during the COVID-19 pandemic and protests; it then increased through August, culminated in a peak during September amid intensified pre-election campaigning, and experienced a modest decline in October. This variability underscores the responsiveness of political discourse to legislative schedules and escalating election pressures, as visualized through monthly aggregations that affirm the period's overall volatility.

In addressing RQ3, the Engagement Myth—assessing whether sentiment polarity correlates significantly with engagement metrics like likes and retweets, or if other variables predominate—the findings revealed

no substantial link, with correlations near zero ( $\sim 0.00$  for retweets and  $\sim 0.07$  for likes). This challenges the notion that emotional intensity drives virality; instead, factors such as account prominence (e.g., follower base) and topical relevance (e.g., alignment with current events) emerge as the key influencers. Thus, heightened positivity or negativity does not inherently enhance interaction, highlighting that the communicator's identity and the content's substance outweigh affective tone in determining propagation.

## V. SUMMARY

### **Was the Purpose Achieved?**

The project's purpose was indeed fully realized through a structured approach to sentiment classification, methodological comparison, and pattern identification in political tweets. By systematically applying diverse NLP techniques, it highlighted the strengths of transformer models relative to conventional methods, effectively transferred sophisticated predictions to a streamlined machine learning setup, and revealed important aspects of political discourse, including partisan variations and the limited influence of sentiment on engagement levels. This accomplishment not only affirmed the utility of NLP for handling complex social media data but also yielded valuable implications for media oversight and communication tactics, as demonstrated by the detailed results and interpretations.

With respect to the first objective—conducting multi-method sentiment labeling to categorize tweets as Negative, Neutral, or Positive using dictionary-based, transformer, and supervised machine learning approaches—the goal was met successfully. All 30,490 tweets underwent labeling across the various techniques, generating separate CSV files with varying distributions (for instance, VADER classified 64.7% as Positive, the transformer identified 49.6% as Neutral, and logistic regression yielded 40.5% Positive), thereby proving the methods' viability in the face of challenges such as hashtags and informal language.

The second objective—undertaking a comparative methodological analysis to assess performance, biases, and outputs—was likewise achieved. Through examinations of distributions and concordance metrics (e.g., 98.7% agreement between the transformer and logistic regression, contrasted with 58.5% between VADER and the transformer), the study exposed biases in dictionary approaches, such as their tendency to overestimate positivity, while underscoring the transformer's edge in contextual interpretation; the logistic regression model, in turn, attained 81.22% accuracy against silver labels.

Finally, the third objective—discovering and analyzing patterns related to party affiliation, temporal dynamics, and engagement—was fulfilled effectively. The analysis disclosed a notable positivity disparity among Republicans (46.5%) compared to Democrats (35.9%), along with time-based shifts (e.g., a dip in June-July followed by a September high) and negligible correlations between sentiment and engagement ( $\sim 0.00$  for retweets), attributing greater predictive power to factors like account influence and subject matter relevance.

## Were the Assumptions Examined?

The assumptions underpinning the project were examined to varying degrees. While some were directly tested through comparative analysis and statistical evaluation, others received only partial scrutiny due to the lack of human-annotated ground truth and external validation.

The assumption that sentiment in political tweets can be reliably captured by automated models without human-annotated ground truth, relying instead on "silver labels" from advanced models, was examined only partially. This assumption was indirectly tested through the high agreement (98.7%) between the transformer and logistic regression models, which confirmed that silver labels serve as a viable proxy. However, the absence of human validation was explicitly acknowledged as a key limitation, noting that agreement metrics do not guarantee absolute accuracy.

The second assumption—that transformer models such as RoBERTa provide sufficiently context-aware and trustworthy predictions to serve as pseudo-ground truth—was examined in full. Comparative analysis of sentiment distributions and model agreement rates demonstrated clear superiority over dictionary methods: RoBERTa identified 49.6% neutral tweets compared to VADER's 20.0%, substantially reducing the optimism bias and validating its greater reliability for political text.

The assumption that the dataset is representative of broader political communication styles during the selected period (May–October 2020) was only partially examined. Temporal trends and partisan patterns were analyzed, and sentiment dips (e.g., June–July lull) were linked to major events such as COVID-19, pre-election events, and protests. Nevertheless, the assumption was not deeply challenged through external validation or comparison with broader datasets, leaving potential selection or temporal biases unaddressed.

The assumption that engagement metrics are primarily influenced by sentiment, party affiliation, or timing (rather than other factors) was directly tested and largely refuted. Correlation analysis revealed near-zero relationships ( $\approx 0.00$  for retweets and  $\approx 0.07$  for likes), indicating that sentiment plays a negligible role. Instead, account prominence and topical relevance emerged as stronger predictors of virality.

Finally, the assumption that political language, despite its slogans and nuances, follows patterns detectable by standard NLP methods without extensive domain-specific adjustments, was thoroughly examined and confirmed. Biases in dictionary approaches (e.g., VADER's 64.7% positivity rate due to slogan misinterpretation) clearly demonstrated the limitations of lexicon-based methods, while RoBERTa's superior context handling supported the conclusion that more advanced NLP techniques are essential for this domain.

## Conclusions

The analysis yields several key conclusions regarding the efficacy of NLP methodologies and patterns in political discourse. Transformer models demonstrate superior performance over dictionary-based approaches in analyzing political text, primarily by incorporating contextual nuances and identifying neutrality, thereby mitigating the optimism bias inherent in slogans; consequently, the study advocates for the abandonment of simplistic lexicons in favor of more advanced techniques for domains characterized by rhetorical complexity.

Furthermore, the application of knowledge distillation through silver labels facilitates the development of efficient classical machine learning models, achieving an 81.22% accuracy on the holdout test set and offering scalable solutions for ongoing monitoring of social media sentiment. This conclusion stems from the supervised ML evaluation and silver label strategy, where logistic regression attained 81.22% accuracy alongside 98.7% agreement with the transformer outputs, underscoring the viability of knowledge transfer for rapid, large-scale implementations.

In terms of partisan dynamics, Republicans tend to employ more positive language (46.5% positive sentiment) to mobilize support, whereas Democrats emphasize critiques, resulting in lower positivity (35.9%) and higher negativity (13.8%); these disparities, derived from the partisan divide analysis (RQ1), reflect deliberate strategic choices in communication rather than underlying personal sentiments, with consistency observed across all evaluated methods.

Sentiment also exhibits fluctuations tied to external events, such as dips during the June-July period and peaks in pre-election September, yet it shows no significant influence on engagement metrics; virality is instead driven by communicator identity and topical relevance, as evidenced by near-zero correlations ( $\sim 0.00$  for retweets and  $\sim 0.07$  for likes) from the temporal trends and engagement analysis (RQ2-3), with monthly variations attributing shifts to factors like legislative calendars and campaign intensity.

Overall, the investigation affirms the value of NLP in political analysis while identifying critical areas for improvement, including the integration of human annotation to establish ground truth and enhanced mechanisms for detecting sarcasm; these insights, drawn from the critical review of limitations and potential applications, highlight the absence of human validation as a primary constraint and recommend future efforts toward event-specific correlations to bolster analytical precision.

## VI. APPENDIX

**Source Code:** Available in the GitHub repository: [GitHub](#). Includes three Jupyter notebooks (`senator_sentiment_dictionary.ipynb`, `senator_sentiment_transformers.ipynb`, `senator_sentiment_TF-IDF+LR+CV.ipynb`) for end-to-end implementation, using libraries like Pandas, Scikit-Learn, Hugging Face Transformers, and VADER.

**Data Set:** Primary file `senator_twitter_May-Oct.csv` (from Harvard Dataverse: [Harvard.edu](#)). Output CSVs with labels: `senator_tweets_with_sentiment_plus_vader.csv`, `senator_tweets_with_sentiment.csv`, `senator_tweets_with_lr_tfidf.csv`.